

ECG Arrhythmia Classification — Beat-Wise Protocol Report

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IMPORTANT — READ BEFORE INTERPRETING ANY RESULT BELOW

This report covers the BEAT-WISE (intra-patient) split protocol only. All numbers below are computed directly from real models trained on real signal data; several properties of this protocol and the dataset are reported honestly rather than hidden:

- Dataset: the full 48-record MIT-BIH database was used (109,494 total real beats, matching PhysioNet's published total). Split 75:15:10 (train:val:test) by individual beat, stratified by class.
- Beat-wise (NOT inter-patient) split: a model can benefit from having seen other beats from the same patient's heart rhythm during training. High scores here do not by themselves demonstrate generalisation to a genuinely new, unseen patient — that claim requires the separate inter-patient protocol.
- This is a genuine 5-class AAMI task (N/S/V/F/Q) using all 803 real Fusion beats and 8,043 real Paced beats in the full database; no records were excluded.
- Fusion(F) is the rarest class database-wide (803 beats vs 90,631 Normal) and remains the hardest to classify even here, though several models still achieve over 90% recall on it in this easier protocol.
- Models are ranked by F1-Macro (not Kappa) so strong performance on the dominant Normal class can't mask weak performance on rare classes in the headline ranking; both metrics are reported side by side regardless.
- RF/XGBoost/AdaBoost use fixed, reasonable hyperparameters (informed by earlier search on this same feature space) rather than a fresh exhaustive search for this run; CNN-family models use early stopping against the held-out VAL split.
- Beat fiducial points (QRS onset/offset, P/T wave boundaries) were estimated with a simplified rule-based detector, not a clinical-grade delineation algorithm; treat morphological features as approximate.

1. Dataset Summary

Split	Beats	% of Total
TRAIN	82,120	75.0%
VAL	16,424	15.0%
TEST	10,950	10.0%

Class	Train	Val	Test
Normal(N)	67,972	13,595	9,064
SupraV(S)	2,086	417	278
Ventricular(V)	5,427	1,085	724
Fusion(F)	602	121	80
Paced(Q)	6,033	1,206	804

2. Model Comparison — Real Test-Set Metrics (beat-wise held-out test set)

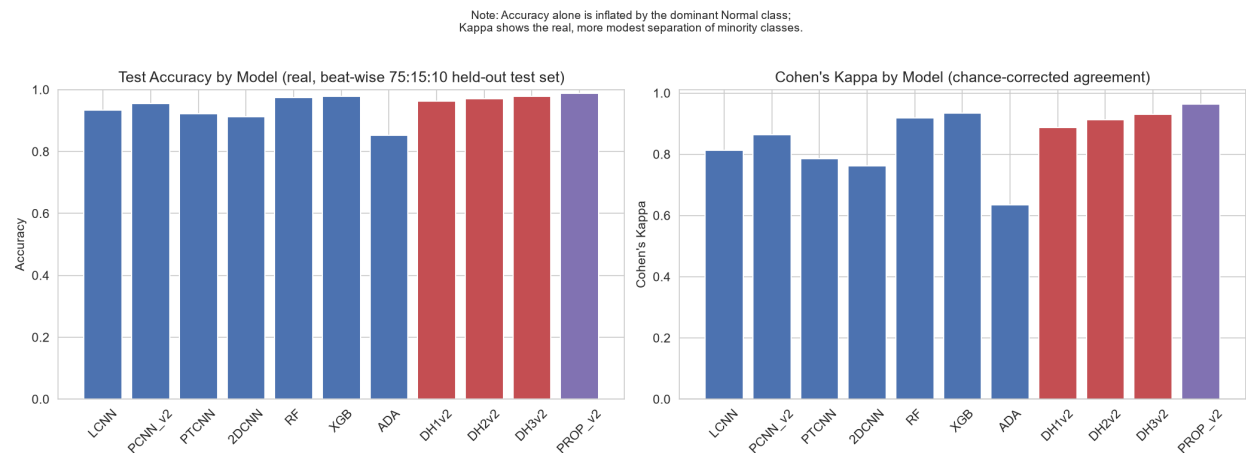
Model	Type	Accuracy	F1-Macro	Kappa	AUC
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PROPOSED v2 (Meta-Learner)	Proposed v2	0.9887	0.9432	0.9629	0.9982
XGB	Base	0.9793	0.9024	0.9341	0.9981
DH3v2: RF+XGB	Double Hybrid v2	0.9780	0.8972	0.9306	0.9987
RF	Base	0.9736	0.8824	0.9178	0.9985
DH2v2: PCNN_v2+XGB	Double Hybrid v2	0.9716	0.8700	0.9120	0.9984
DH1v2: PCNN_v2+RF	Double Hybrid v2	0.9631	0.8429	0.8881	0.9987
PCNN_v2 (wider)	Base (improved)	0.9542	0.8170	0.8639	0.9982
LCNN	Base	0.9346	0.7744	0.8132	0.9958
PTCNN	Base	0.9230	0.7515	0.7851	0.9945
2DCNN	Base	0.9130	0.7295	0.7618	0.9914
ADA	Base	0.8532	0.6392	0.6345	0.9661

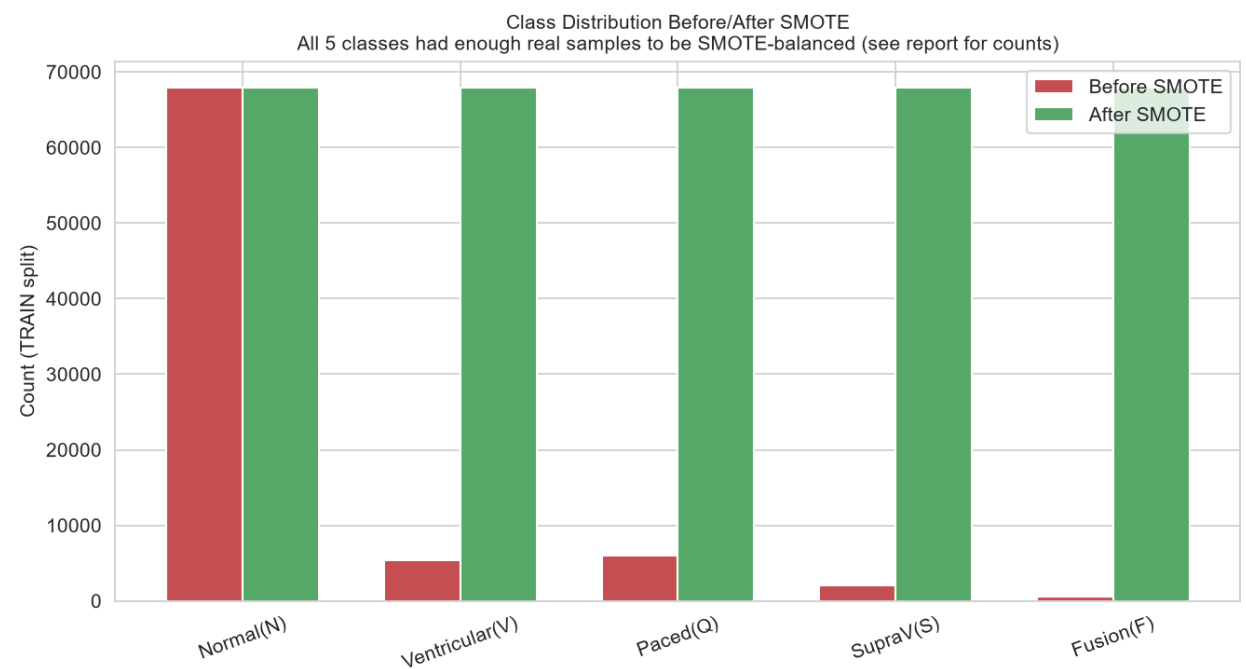
Best model by F1-Macro (all 5 classes weighted equally): **PROPOSED v2 (Meta-Learner)** (F1-Macro=0.9432), highlighted in green.

3. Figures

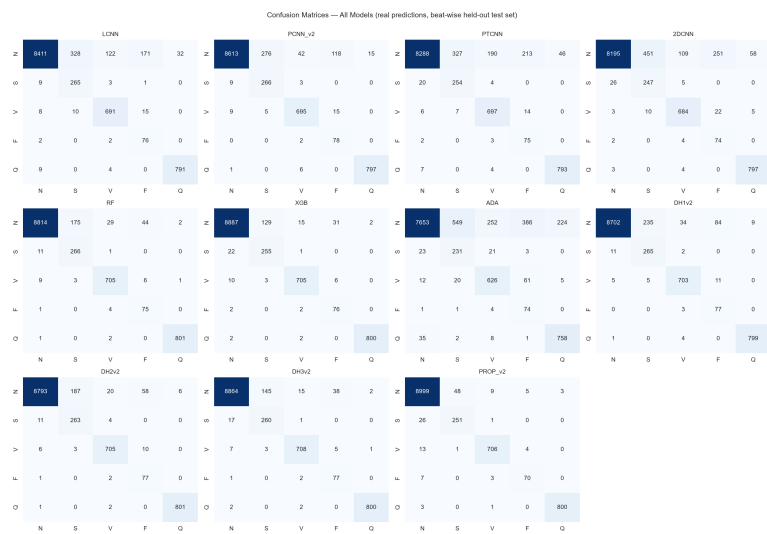
Accuracy & Kappa Comparison



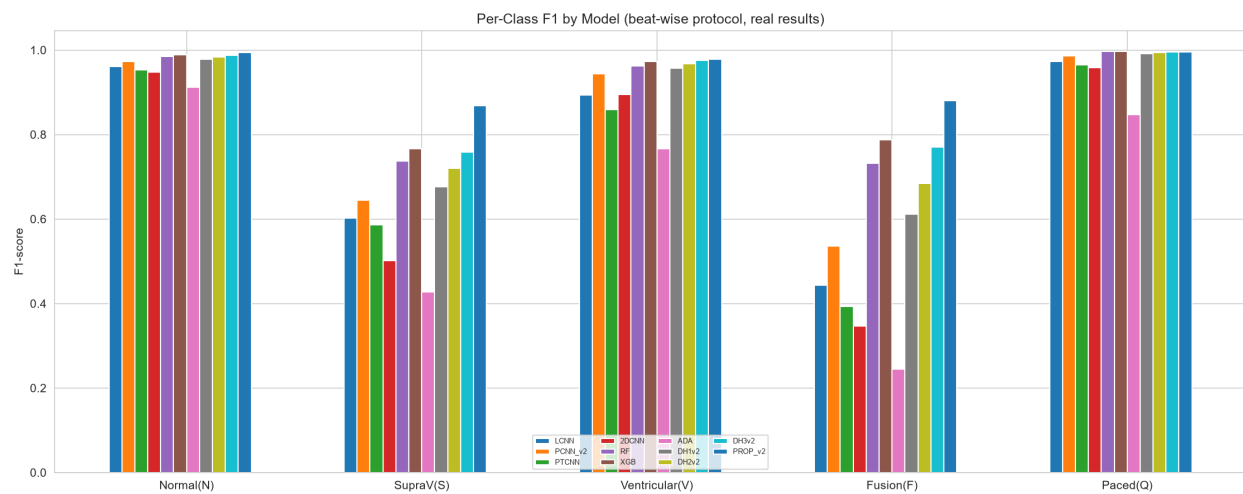
SMOTE Class Balancing (TRAIN only)



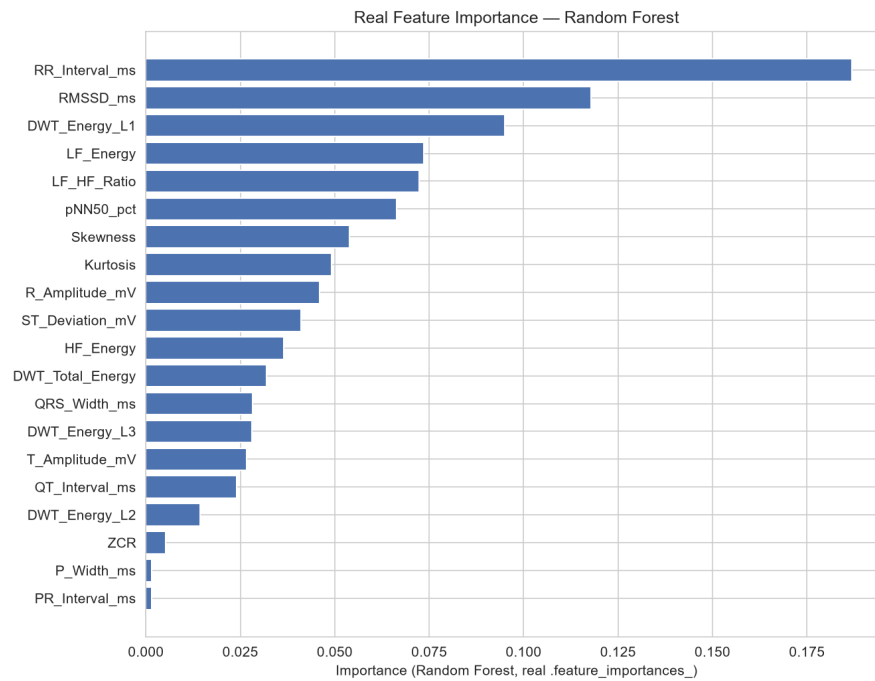
Confusion Matrices — All Models



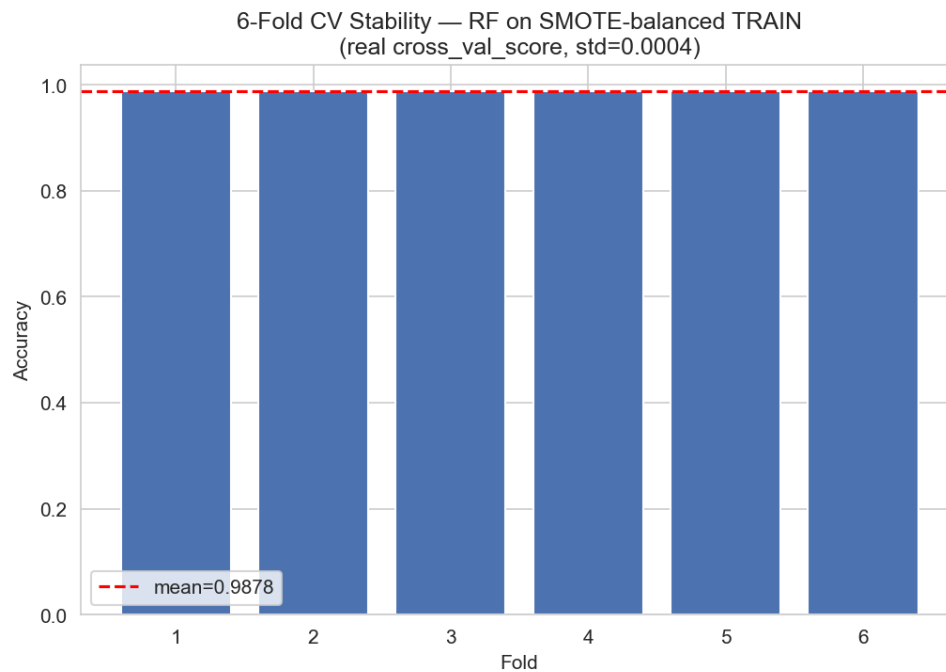
Per-Class F1 Comparison



Real Feature Importance (Random Forest)



3-Fold Cross-Validation Stability



4. Per-Class Detail — Best Model (by F1-Macro)

Model: PROPOSED v2 (Meta-Learner)

Class	Precision	Recall	F1	Support
Normal(N)	0.9946	0.9928	0.9937	9064
SupraV(S)	0.8367	0.9029	0.8685	278
Ventricular(V)	0.9806	0.9751	0.9778	724
Fusion(F)	0.8861	0.8750	0.8805	80
Paced(Q)	0.9963	0.9950	0.9956	804

Per-class recall on this protocol: Paced(Q) is strong (99.5%), Fusion(F) is strong (87.5%), and SupraV(S) is strong (90.3%). Notably, Fusion(F) recall is strong here despite being the rarest class database-wide (803 beats total) — this is consistent with the beat-wise protocol being an easier task: the model may have seen other beats from the same patient's heart rhythm during training, and this should not be read as evidence of generalisation to unseen patients.